# SALES PREDICTION USING PYTHON

## A Machine Learning Regression Project

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**Programme:** CodeAlpha Data Science Internship
**Project:** Sales Prediction Using Python
**Dataset:** Advertising.csv
**Selected Model:** Tuned Gradient Boosting Regressor
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## EXECUTIVE SUMMARY

Sales prediction is an important application of data science that helps organizations understand historical sales patterns and make informed business decisions. This project applies machine learning regression techniques to predict sales using advertising expenditure across three media channels: television (TV), radio, and newspaper.

The project follows a structured data science workflow, including dataset auditing, data cleaning, exploratory data analysis (EDA), feature preparation, model development, cross-validation, hyperparameter tuning, and final model evaluation.

Several regression algorithms were evaluated to identify a model capable of accurately predicting sales. Following model comparison and hyperparameter tuning, the **Tuned Gradient Boosting Regressor** was selected as the final model.

The model achieved the following results on a held-out test set of 40 observations:

- **Mean Absolute Error (MAE):** 0.6481
- **Root Mean Squared Error (RMSE):** 0.7935
- **Coefficient of Determination ($R^2$):** 0.9801

The $R^2$ score indicates that the model explains approximately 98.01% of the variation in sales within the test sample relative to a mean-only prediction baseline.

These results demonstrate strong predictive performance on the evaluated dataset. However, additional validation is recommended before using the model for operational forecasting or advertising budget decisions. Predictive relationships should not automatically be interpreted as evidence that increasing advertising expenditure will cause sales to increase.

## 1. INTRODUCTION

### 1.1 Background

Businesses invest in advertising to promote products, reach customers, and increase sales. Advertising budgets may be distributed across multiple channels, including television, radio, and newspapers.

Determining how advertising expenditure relates to sales can help businesses evaluate historical campaign performance and improve planning. Data science and machine learning provide techniques for identifying patterns in historical observations and using those patterns to estimate future or unseen outcomes.

Regression is particularly appropriate when the objective is to predict a continuous numerical variable, such as sales.

This project uses the Advertising.csv dataset to develop and evaluate machine learning regression models for predicting sales from advertising expenditure.

### 1.2 Problem Statement

Organizations may find it difficult to estimate sales outcomes from advertising expenditure using manual calculations or simple assumptions alone.

A data-driven approach can help identify relationships between advertising channels and observed sales. However, the usefulness of such an approach depends on the quality of the data, the selected algorithm, and the reliability of the model evaluation process.

This project addresses the problem by comparing regression algorithms and selecting a model based on its predictive performance.

### 1.3 Project Objectives

The main objective is to develop a machine learning model that predicts sales from advertising expenditure.

The specific objectives are to:

1. Examine the structure and quality of the Advertising.csv dataset.
2. Identify missing values, duplicate records, inappropriate columns, and unusual observations.
3. Perform exploratory data analysis to understand the distributions and relationships among variables.
4. Prepare the dataset for supervised machine learning.
5. Train and compare different regression algorithms.
6. Apply cross-validation and hyperparameter tuning to improve model performance.
7. Evaluate the final model using MAE, RMSE, and $R^2$.
8. Interpret the findings and provide recommendations for further analysis and business planning.

### 1.4 Scope of the Project

The project focuses on predicting sales using three advertising expenditure variables:

- TV advertising expenditure
- Radio advertising expenditure
- Newspaper advertising expenditure

The target variable is Sales.

The project covers data preparation, exploratory analysis, regression modelling, evaluation, and interpretation. It does not establish causal relationships between advertising expenditure and sales.

## 2. DATASET DESCRIPTION

### 2.1 Dataset Overview

The project uses Advertising.csv, a dataset containing advertising expenditure and sales observations.

The source dataset contains **200 rows and 5 columns**, including an index-like column.

| Column | Role | Description |
|---|---|---|
| Unnamed: 0 | Index | Sequential identifier that is not intended to be a predictive feature |
| TV | Predictor | Television advertising expenditure |
| Radio | Predictor | Radio advertising expenditure |
| Newspaper | Predictor | Newspaper advertising expenditure |
| Sales | Targe | Observed sales value to be predicted |

The three advertising expenditure variables serve as input features, while Sales is the output variable.

The exact measurement units should be confirmed from the dataset documentation before being specified in a final publication.

### 2.2 Type of Machine Learning Problem

This project is a **supervised machine learning regression problem**.

Supervised learning uses examples containing input variables and known target values to learn a predictive relationship.

Regression is suitable because Sales is a continuous numerical variable rather than a discrete class label.

The modelling problem can be represented as:

Sales = f(TV, Radio, Newspaper)

Here, the function f represents the relationship learned by the machine learning algorithm.

### 2.3 Feature Selection

The TV, Radio, and Newspaper variables were selected as predictors because they represent the advertising channels being investigated.

The Unnamed: 0 column represents a sequential index rather than a meaningful advertising characteristic. It should therefore be excluded from the model's feature matrix.

The final modelling structure is:

- **Input features (X):** TV, Radio, Newspaper
- **Target variable (y):** Sales

## 3. DATA AUDITING AND PREPROCESSING

### 3.1 Data Auditing

Data auditing was conducted to understand the dataset's structure and identify potential quality problems before modelling.

The audit covered:

- Dataset dimensions
- Column names and data types
- Descriptive statistics
- Missing values
- Duplicate records
- Negative or unusual values
- The index-like column

The recorded audit findings indicated that the dataset contained no missing values and no duplicate records. The substantive advertising and sales fields also contained no negative values.

These checks reduced the need for extensive missing-value imputation or duplicate removal.

### 3.2 Handling the Index Column

The column named Unnamed: 0 behaves like a sequential row identifier. Such columns can introduce irrelevant information if treated as model predictors.

It was therefore identified for exclusion from the feature matrix.

Removing the index-like column helps ensure that the model learns from the advertising variables rather than relying on row order.

### 3.3 Outlier Analysis

The Interquartile Range (IQR) method was used to flag potentially unusual observations.

The IQR is calculated as:

$$IQR = Q3 - Q1$$

Where:

- Q1 is the first quartile.
- Q3 is the third quartile.

The conventional IQR boundaries are:

$$\text{Lower boundary} = Q1 - 1.5 \times IQR$$

$$\text{Upper boundary} = Q3 + 1.5 \times IQR$$

Observations outside these boundaries may be flagged for further investigation.

Some Newspaper expenditure values, including 100.9 and 114.0, were identified as high values relative to the distribution.

These observations were not automatically treated as errors. An unusual observation may represent a genuine advertising expenditure amount rather than incorrect data.

Retaining plausible observations avoids unnecessary information loss. However, the influence of unusual values should be assessed through visual inspection, domain knowledge, and model diagnostics.

### 3.4 Data Preparation Summary

The preparation process involved the following steps:

1. Load the dataset into Python using Pandas.
2. Inspect the dimensions, column names, data types, and descriptive statistics.
3. Check for missing values and duplicate records.
4. Investigate unusual observations.
5. Exclude the index-like column from predictive features.
6. Separate the input variables from the target variable.
7. Divide the observations into training and test sets.

A holdout test set of 40 observations was used for the reported final evaluation.

## 4. EXPLORATORY DATA ANALYSIS

### 4.1 Purpose of EDA

Exploratory Data Analysis (EDA) was used to understand the structure, distributions, and relationships in the dataset before building predictive models.

EDA helps identify patterns, potential anomalies, and relationships that may influence model selection and interpretation.

### 4.2 Descriptive Statistics

Descriptive statistics provide an overview of the distribution of each numerical variable.

Important statistics include:

- Mean
- Median
- Standard deviation
- Minimum and maximum values
- First and third quartiles

These measures help describe typical advertising expenditure and sales levels, as well as the variability present in the data.

### 4.3 Distribution Analysis

Histograms and boxplots are useful for examining the distributions of TV, Radio, Newspaper, and Sales.

Histograms reveal the frequency distribution of observations, while boxplots help identify skewness and potentially unusual values.

Such visualizations are particularly useful when deciding whether data transformations or additional investigation are necessary.

### 4.4 Relationship Analysis

Scatterplots can be used to compare each advertising channel with Sales.

The analysis should consider:

- Whether the relationship appears positive or negative.
- Whether the pattern is approximately linear or non-linear.
- Whether observations appear to form clusters.
- Whether unusual observations may influence the apparent relationship.

A correlation matrix can also summarize the direction and strength of linear associations between the numerical variables.

However, correlation does not establish causation. Other factors not represented in the dataset may influence sales.

### 4.5 EDA Interpretation

The EDA stage provides the foundation for selecting and evaluating predictive models.

For a complete project submission, the final notebook should retain the actual EDA figures and numerical outputs. Any specific claims about which advertising channel has the strongest relationship with sales should be supported by the computed correlations, plots, or model analysis rather than assumed in advance.

## 5. MACHINE LEARNING METHODOLOGY

### 5.1 Model Selection

Several regression algorithms were considered to compare their ability to predict sales.

The model comparison process included baseline regression approaches and tree-based ensemble methods.

The following algorithms are relevant to the project:

**Linear Regression:** Establishes a simple baseline by modelling a linear relationship between the predictors and sales.

**Decision Tree Regressor:** Learns decision rules that divide the feature space into regions with different predicted sales values.

**Random Forest Regressor:** Combines predictions from multiple decision trees to improve robustness and capture non-linear relationships.

**Gradient Boosting Regressor:** Builds an ensemble sequentially, with each new learner attempting to improve the predictions made by the existing ensemble.

The final selected algorithm was the Tuned Gradient Boosting Regressor.

### 5.2 Why Gradient Boosting Was Selected

Gradient Boosting was selected because it achieved strong predictive performance after hyperparameter tuning.

The method combines multiple weak learners, commonly decision trees, into a more powerful predictive model.

Its advantages include:

1. Ability to model non-linear relationships.
2. Ability to capture interactions between input features.
3. Flexible control over model complexity through hyperparameters.
4. Potential to achieve high predictive accuracy on structured tabular datasets.

The model's performance in this project supports its selection for the evaluated dataset. Nevertheless, the final choice should also consider validation stability, interpretability, and the performance of simpler alternatives.

### 5.3 Training and Testing

The dataset was divided into training and test subsets.

The training data were used to fit the candidate models and conduct model selection. The test data were reserved for final evaluation.

The reported test set contained 40 observations.

Keeping the test set separate from model tuning helps provide a more independent estimate of predictive performance.

### 5.4 Cross-Validation and Hyperparameter Tuning

Cross-validation helps estimate model performance across multiple training and validation partitions.

Five-fold cross-validation was used during the model-selection and tuning process.

Hyperparameter tuning was performed to identify model configurations that improved validation performance.

The Gradient Boosting model was tuned before its final evaluation.

For reproducibility, the exact best hyperparameters, random seed, cross-validation score, and package versions should be recorded from the final executed notebook.

## 6. MODEL EVALUATION

### 6.1 Evaluation Metrics

Three metrics were used to evaluate the selected model: Mean Absolute Error, Root Mean Squared Error, and the coefficient of determination.

**Mean Absolute Error (MAE)**

MAE measures the average absolute difference between actual and predicted values.

$$MAE = (1/n) \times \Sigma |y_i - \hat{y}_i|$$

A lower MAE indicates that predictions are, on average, closer to the actual values.

**Root Mean Squared Error (RMSE)**

RMSE measures the square root of the average squared prediction errors.

$$RMSE = \sqrt{[(1/n) \times \Sigma (y_i - \hat{y}_i)^2]}$$

Because errors are squared, RMSE gives greater weight to larger prediction errors.

A lower RMSE generally indicates better predictive performance.

**Coefficient of Determination ($R^2$)**

$R^2$ compares the model's prediction errors with the variation in the target variable around its mean.

$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$

An $R^2$ of 1 represents a perfect fit on the evaluated observations. An $R^2$ of 0 indicates performance equivalent to predicting the test-set mean under the standard definition, while negative values indicate worse performance than that baseline.

### 6.2 Final Model Results

The Tuned Gradient Boosting Regressor achieved the following results:

| Evaluation metric | Result |
|---|---|
| Selected model | Gradient Boosting (Tuned) |
| Test MAE | 0.6481 |
| Test RMSE | 0.7935 |
| Test $R^2$ | 0.9801 |
| Test samples | 40 |

### 6.3 Interpretation of Results

**MAE = 0.6481**

The model's predictions differed from actual sales values by an average absolute amount of approximately 0.6481 target units on the test set.

The practical meaning of this error depends on the units used in the original dataset.

**RMSE = 0.7935**

The RMSE indicates that the model achieved a relatively low squared-error-based prediction loss on the evaluated observations.

Since RMSE penalizes larger errors more strongly than MAE, comparing the two metrics can help reveal whether a small number of large errors may be affecting performance.

**$R^2$ = 0.9801**

The $R^2$ value indicates that the model explained approximately 98.01% of the variation in sales within the held-out test sample relative to a mean-only baseline.

This is a strong result for the evaluated split. It does not guarantee that the model will achieve the same performance on different samples or future business data.

### 6.4 Overall Model Assessment

The reported metrics indicate that the Tuned Gradient Boosting Regressor performed well on the 40-observation test set.

The model is a promising candidate for further investigation and potential use in sales prediction.

However, the dataset contains only 200 observations. Repeated evaluation, independent validation, and ongoing monitoring are recommended before relying on the model in a production environment.

## 7. KEY FINDINGS

The project supports the following findings:

1. The dataset provides advertising expenditure variables that can be used to predict observed sales.
2. Data auditing helps identify irrelevant columns and unusual values before model training.
3. Outlier detection should be followed by investigation rather than automatic deletion.
4. Comparing multiple regression algorithms provides a more informed basis for selecting a final model.
5. Hyperparameter tuning can improve model performance when evaluated using an appropriate validation strategy.
6. The Tuned Gradient Boosting Regressor achieved an MAE of 0.6481, RMSE of 0.7935, and $R^2$ of 0.9801 on the reported test set.
7. High predictive accuracy does not, by itself, demonstrate that changing advertising expenditure will cause a corresponding change in sales.

## 8. BUSINESS IMPLICATIONS

### 8.1 Sales Planning

A validated model could help businesses estimate sales outcomes for advertising expenditure patterns similar to those represented in the training data.

Predictions may support preliminary planning, comparison of scenarios, and identification of observations that warrant further investigation.

### 8.2 Advertising Performance Analysis

The relationship between each advertising channel and sales can be explored through visualizations and model interpretation techniques.

This may help analysts identify patterns worth investigating in campaign performance.

However, predictive feature importance should not be interpreted directly as the causal return on investment of an advertising channel.

### 8.3 Decision Support

Businesses could use model predictions as one input into planning discussions.

Decisions should also consider campaign objectives, advertising costs, customer behaviour, timing, competition, and uncertainty in the predictions.

A predictive model should support human decision-making rather than replace business judgment.

## 9. LIMITATIONS

Several limitations should be considered when interpreting the results.

### 9.1 Small Dataset

The dataset contains 200 observations. This limits the range of conditions the model can learn and makes performance estimates potentially sensitive to the train/test split.

### 9.2 Limited Predictors

The model uses only TV, Radio, and Newspaper advertising expenditure.

Other factors that may influence sales, such as pricing, seasonality, product characteristics, promotions, customer demographics, and competitor activity, are not represented in these features.

### 9.3 Observational Data

The dataset describes observed relationships. It does not establish that advertising expenditure independently causes sales changes.

Causal claims would require a suitable experimental or causal-inference design.

### 9.4 Validation Stability

The reported metrics come from a single 40-observation test set.

Performance may differ under another split or on data collected under different conditions. Repeated validation would help assess stability.

### 9.5 Generalization to New Environments

The model may perform less effectively if future advertising patterns or market conditions differ substantially from those represented in the dataset.

### 9.6 Interpretability

Gradient Boosting can be more difficult to explain than a simple linear model. Additional diagnostic techniques may be necessary to communicate how predictions are produced.

## 10. RECOMMENDATIONS AND FUTURE WORK

The following improvements are recommended:

1. **Perform repeated validation.** Evaluate the model across multiple splits or repeated cross-validation to assess the stability of the results.

2. **Compare with simpler baselines.** Retain Linear Regression and a mean-only baseline to determine whether the additional model complexity provides meaningful benefits.
3. **Record the final hyperparameters.** Include the exact best parameters and cross-validation results in the project repository.
4. **Investigate model behaviour.** Use permutation importance, partial dependence, or other suitable interpretation techniques to understand predictive patterns.
5. **Expand the dataset.** Incorporate more observations and, where possible, additional relevant business features.
6. **Use time-aware validation where appropriate.** If time-stamped campaign data become available, evaluate the model using a validation strategy that respects chronological order.
7. **Monitor model performance.** If the model is deployed, compare predictions with subsequently observed sales and investigate changes in error.
8. **Investigate causal effects separately.** Use controlled experiments or an appropriate causal-inference framework before recommending specific advertising budget changes on causal grounds.
9. **Improve reproducibility.** Maintain a requirements.txt file, a clear README, saved figures, documented model parameters, and a reproducible training script or notebook.

## 11. CONCLUSION

This project applied machine learning regression techniques to sales prediction using TV, Radio, and Newspaper advertising expenditure.

The workflow covered dataset auditing, preprocessing, exploratory data analysis, model comparison, cross-validation, hyperparameter tuning, and final evaluation.

The Tuned Gradient Boosting Regressor was selected as the final model. It achieved an MAE of 0.6481, an RMSE of 0.7935, and an $R^2$ of 0.9801 on a held-out test set containing 40 observations.

These results indicate strong predictive performance on the evaluated dataset. The project demonstrates how a structured data science workflow can be used to build, compare, and evaluate predictive models.

Further validation and richer data are recommended before operational deployment. In particular, predictive accuracy should be distinguished from causal evidence when making advertising and sales decisions.

Overall, the project provides a practical demonstration of supervised machine learning for sales prediction and establishes a foundation for future work in marketing analytics and business intelligence.

## APPENDIX B: PROJECT DETAILS

**Project author:** John Afwilile Kapungu

**Project title:** Sales Prediction Using Python

**Internship programme:** CodeAlpha Data Science Internship

**Dataset:** Advertising.csv

**Selected model:** Tuned Gradient Boosting Regressor

**Final test metrics:**

- MAE: 0.6481
- RMSE: 0.7935
- $R^2$: 0.9801
- Test samples: 40

**GitHub repository:** [Insert repository URL]

**Python version:** python 3.14.7

**Cross-validation score:** [Insert final score and metric]

**Report date:** October 2026